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Pearson Edexcel
Level 1/Level 2 GCSE (9–1)

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Hannah Kettle MATHS TUTORIALS



Half Past Paper – Week 23

Higher Tier

Total Marks

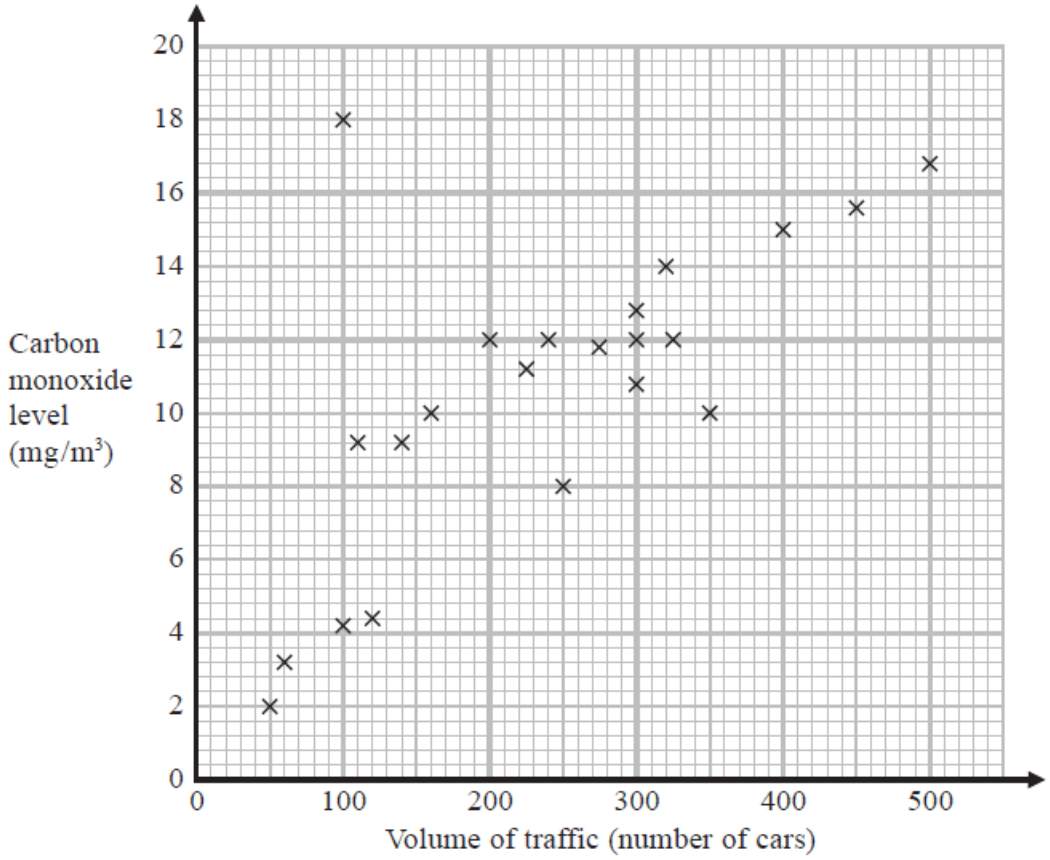
Q	Topic	Marks	Sparx Code
1a	Interpreting scatter graphs	1	U277
1b	Interpreting scatter graphs, Using lines of best fit	2	U277,U128
1c	Interpreting scatter graphs	1	U277
3a	Using standard form with positive indices	1	U330
3b	Using standard form with negative indices	1	U534
3c	Adding and subtracting numbers in standard form	2	U290
5	Special sequences	3	U680
7	Finding the area of sectors	4	U373
9a	Solving inverse proportion word problems	2	U357
9b	Solving inverse proportion word problems	1	U357
11	Finding error intervals	2	U657
13	Constructing and solving simultaneous equations	5	U137
15	Estimating gradients of non-linear graphs	3	U800
17	Constructing direct proportion equations	4	U407
19	Using recurrence relations	3	U171
21a	Estimating areas under non-linear graphs	3	U882
21b	Estimating areas under non-linear graphs	1	U882
23	The sine rule, Graphs of trigonometric functions	4	U952,U450
	Total	43	

Answer ALL questions.

Write your answers in the spaces provided.

You must write down all the stages in your working.

1 The scatter graph shows information about the volume of traffic and the carbon monoxide level at a point on a road each day for 22 days.



One point is an outlier.

(*a*) Write down the coordinates of this point.

(..... , ...)

(1)

For another day, 370 cars pass the point on the road.

(*b*) Estimate the carbon monoxide level for this day.

..... mg/m^3

(2)

Alfie says,

"Because there is an outlier, there is no correlation."

(*c*) Is Alfie correct?
You must give a reason for your answer.

.....

.....

.....

(1)

(Total for Question 1 is 4 marks)

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3 (*a*) Write 4.5×10^5 as an ordinary number.

.....

(1)

(*b*) Write 0.007 in standard form.

.....

(1)

(*c*) Work out $4.2 \times 10^3 + 5.3 \times 10^2$
Give your answer in standard form.

.....

(2)

(Total for Question 3 is 4 marks)

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5 The first four terms of a Fibonacci sequence are

$$a \qquad 2a \qquad 3a \qquad 5a$$

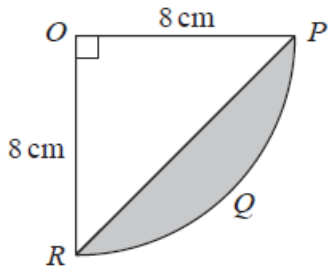
The sum of the first five terms of this sequence is 228

Work out the value of a.

.....

(Total for Question 5 is 3 marks)

7 The diagram shows a sector *OPQR* of a circle, centre *O* and radius 8 cm.



OPR is a triangle.

Work out the area of the shaded segment *PQR*.
Give your answer correct to 3 significant figures.

..... cm^2

(Total for Question 7 is 4 marks)

9 On Monday, 12 people took 5 hours to clean a number of cars.
On Tuesday, 15 people cleaned the same number of cars.

Assuming that all the people worked at the same rate,

(*a*) work out how many hours the 15 people took to clean the cars.

..... hours

(2)

The assumption is wrong.

(*b*) How might this affect the time taken for the 15 people to clean the cars?

.....

.....

(1)

(Total for Question 9 is 3 marks)

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11 Freya writes down the value of x, correct to 1 decimal place.

She writes $x = 6.4$

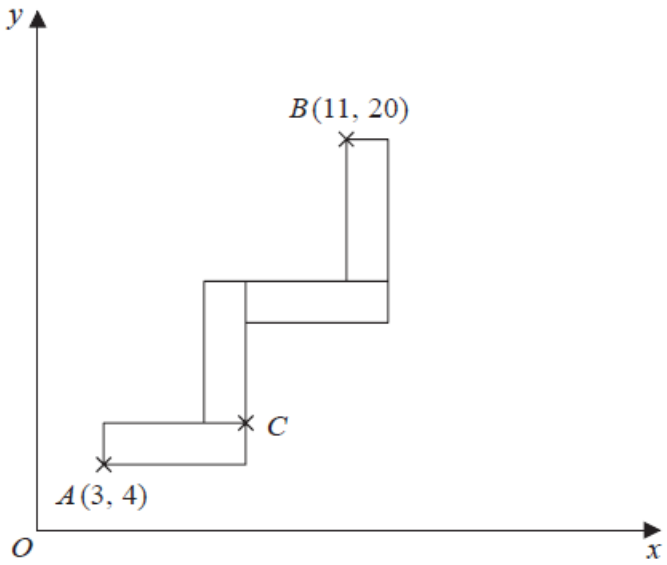
Complete the error interval for x.

..... $\leq x <$ .....

(Total for Question 11 is 2 marks)

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13 A pattern is made from four identical rectangles.
The sides of the rectangles are parallel to the axes.



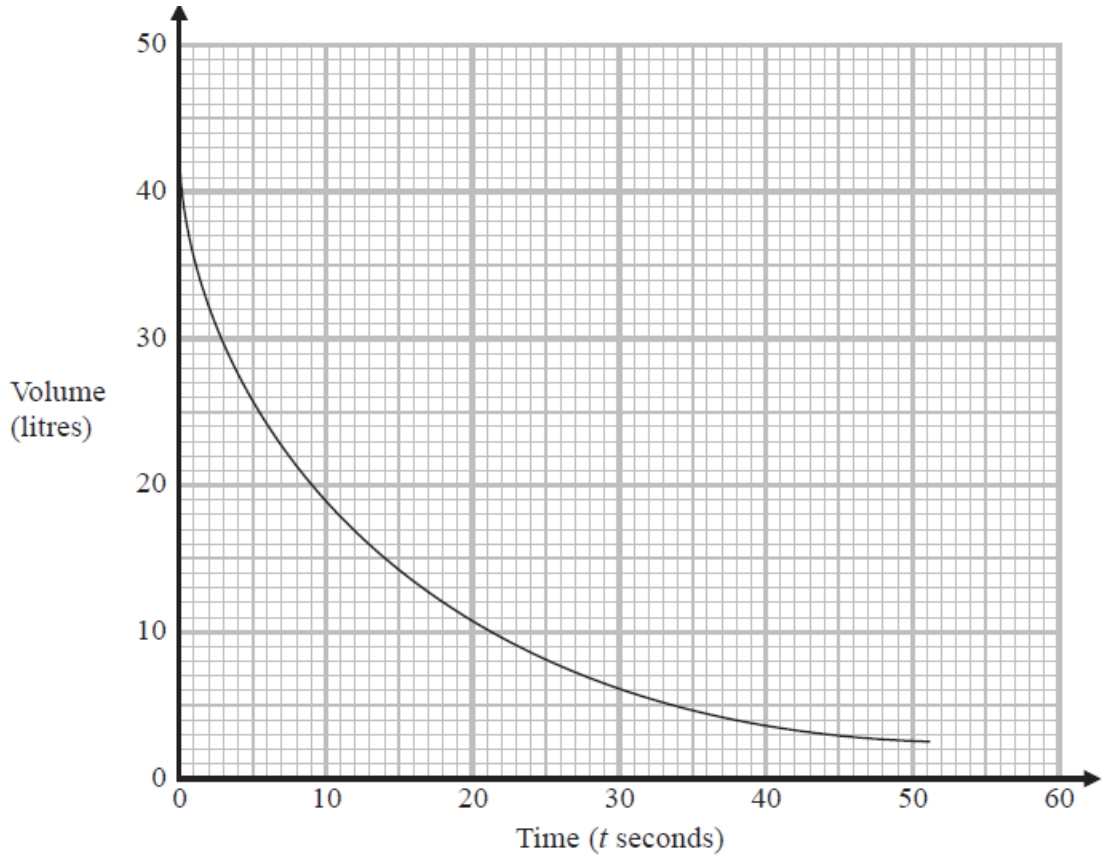
Point *A* has coordinates (3, 4)
Point *B* has coordinates (11, 20)
Point *C* is marked on the diagram.

Work out the coordinates of *C*.
You must show all your working.

(..... , ...)

(Total for Question 13 is 5 marks)

15 The graph gives the volume of water, in litres, in a container at time t seconds after the water started to flow out of the container.



Using the graph, work out an estimate for the rate at which the water is flowing out of the container when $t = 12$
You must show your working.

..... litres per second

(Total for Question 15 is 3 marks)

17 x is directly proportional to the square of y.
y is directly proportional to the cube of z.

$z = 2$ when $x = 32$

Find a formula for x in terms of z.

.....

(Total for Question 17 is 4 marks)

19 At the start of year n, the number of animals in a population is P_n

At the start of the following year, the number of animals in the population is P_{n+1} where

$$P_{n+1} = kP_n$$

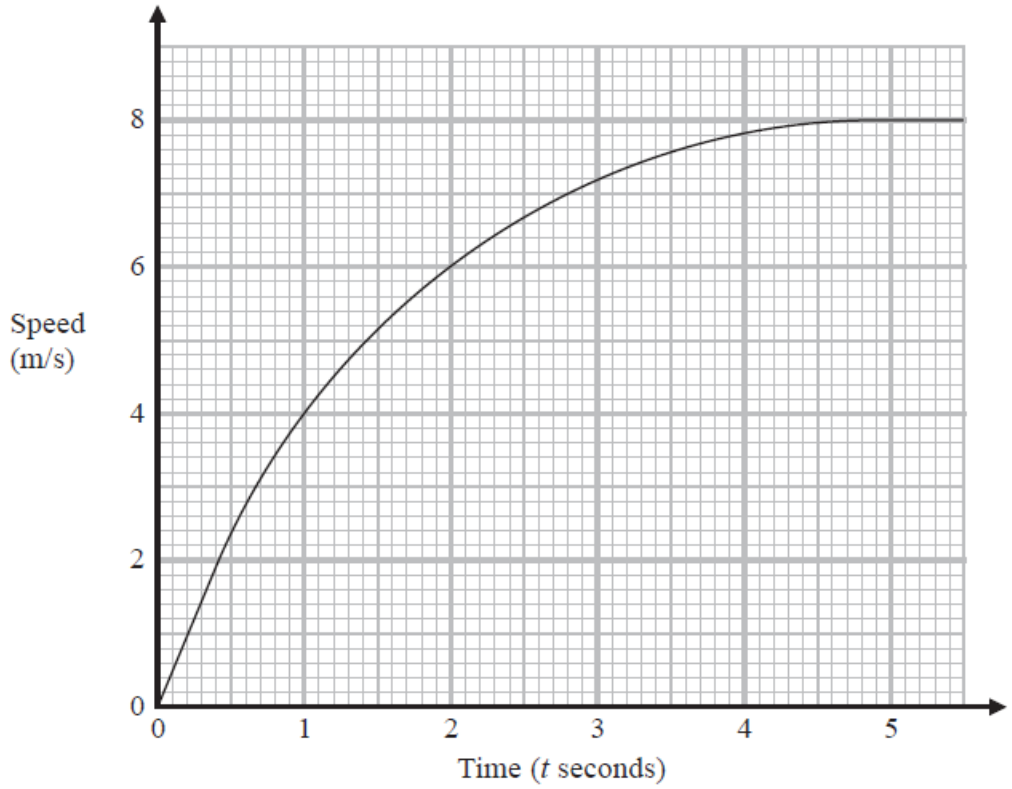
At the start of 2017 the number of animals in the population was 4000
At the start of 2019 the number of animals in the population was 3610

Find the value of the constant k.

.....

(Total for Question 19 is 3 marks)

21 Here is a speed-time graph showing the speed, in metres per second, of an object t seconds after it started to move from rest.



(a) Using 3 trapeziums of equal width, work out an estimate for the area under the graph between $t = 1$ and $t = 4$

.....

(3)

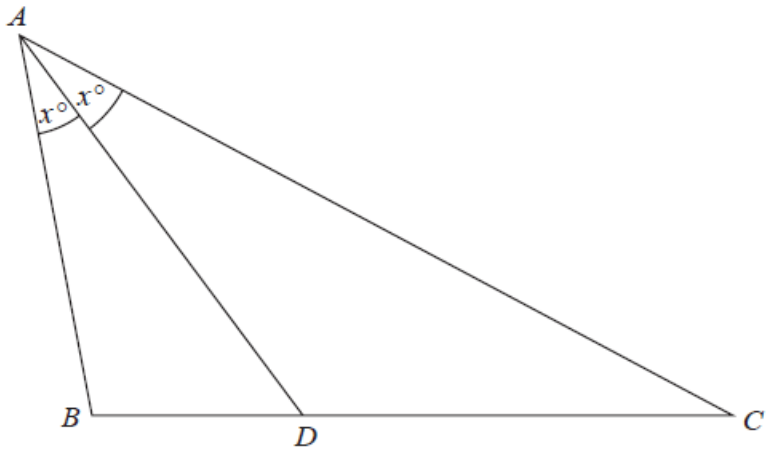
(b) What does this area represent?

.....

(1)

(Total for Question 21 is 4 marks)

23 *ABC* is a triangle.



D is the point on *BC* such that angle *BAD* = angle *DAC* = *x*°

Prove that $\frac{AB}{BD} = \frac{AC}{DC}$

(Total for Question 23 is 4 marks)

TOTAL FOR PAPER IS 43 MARKS